Chapter 2

Sets and Functions

The term 'set' is well known to us. For example, dinner set, set of natural numbers, set of rational numbers etc. As a modern concept set has versatile use. The German mathematician George Cantor (1845-1918) first explained the concept of set. He introduced the notion of infinite set, and created storm in mathematical science. His notion is now known as set theory. In this chapter we shall solve different problems by using concepts of set, diagrams and logic and idea of a function will also be introduced.

After completion of this chapter the learners will —

- be able to express the concepts of sets and subsets through symbols.
- be able to describe how sets can be expressed.
- be able to define infinite sets, and distinguish between finite and infinite sets.
- be able to define union and intersection of sets, and verify.
- be able to define power sets, and construct power sets of sets having two or three elements.
- be able to define ordered pairs and cartesian products.
- be able to prove simple set theoretic rules by using examples and Venn diagrams, and apply these rules for solving different problems.
- be able to define relations and functions, and construct them.
- be able to determine domain and range.
- will be able to draw diagrams for functions.

Sets

A set is simply a well organized object of real or imaginary world. For example, the set of three text books of Bangla, English and Mathematics. Some other

examples are the set of first 10 odd natural numbers, set of integers, set of real numbers etc. Sets are usually denoted by the capital letters of English alphabet like $A, B, C, \ldots X, Y, Z$.

For example, the set of three numbers $2, 4, 6$ can be denoted by $A = \{2, 4, 6\}$

Each object or member of the set is called its element. For example, if $B = \{a, b\}$, to express that a and b are its elements the sign $\in$ is used.

$\therefore a \in B$ and we can read as a, is a member of B (a belongs to B)

$b \in B$ is a member of b, (b belongs to B)

The above set B does not contain c. So it is expressed as $c \notin B$ and is read as c is not a member of B (c does not belong to B)

Methods for expressing sets

Sets can be expressed using two methods, namely Roster method or tabular method and Set Builder Method.

Roster method: In this method all members are listed and put inside {} and if there are more than one element then elements are separated by commas. For example, $A = \{a, b\}$, $B = \{2, 4, 6\}$, $C =$ { Niloy, Shuvro, Tisha } etc.

Set builder method: In this method all elements are not listed rather their common property is described. For example, $A = \{x : x$ is an odd natural number $\}$, $B = \{x : x$ is the first five students of class IX $\}$ etc. Here, by ': has been expressed such that. Since in this method element of the set is specified by rules this method is called **Rule Method**.

Example 1. Express $A = \{7, 14, 21, 28\}$ in set building method.

Solution: The elements of A are $7, 14, 21, 28$.

Here, each element is a multiple of 7 and not exceeding 28.

$\therefore A = \{x : x|$is a multiple of 7 and $0 < x \leq 28\}$

Example 2. Express $B = \{x : x,$ is a divisor of $28\}$ using roster method.

Solution: Here, $28 = 1 \times 28 = 2 \times 14 = 4 \times 7$

$\therefore$ factors of 28 are $1, 2, 4, 7, 14, 28$

Hence the desired set is $B = \{1, 2, 4, 7, 14, 28\}$

Example 3. Express $C = \{x : x \text{ positive integer and } x^2 < 18\}$ using roster method.

Solution: $1, 2, 3, 4, 5, \ldots$ are positive integers.

Here, if

$x = 1$ then, $x^2 = 1^2 = 1$; if $x = 2$ then, $x^2 = 2^2 = 4$

If $x = 3$ then, $x^2 = 3^2 = 9$; if $x = 4$ then, $x^2 = 4^2 = 16$

If $x = 5$ then, $x^2 = 5^2 = 25$; which is greater than 18.

$\therefore$ as per conditions the acceptable positive integers are $1, 2, 3$ and 4.

$\therefore$ the given set is $C = \{1, 2, 3, 4\}$

Work: Express

1) $C = \{-9, -6, -3, 3, 6, 9\}$ using set building method.
2) Express $B = \{y : y \text{ integer and } y^3 \leq 18\}$ using set building method.

Finite Set

The set number of members of which can be determined by counting is called finite set. For example, $D = \{x, y, z\}$, $E = \{3, 6, 9, \ldots, 60\}$, $F = \{x : x \text{ prime and } 30 < x < 70\}$ etc are finite sets. Here, set D contains 3 elements, set E 20 elements and set F 9 elements.

Infinite Set

The set elements of which cannot be exhausted by counting is called infinite set. Fo example, $A = \{x : x \text{ odd natural number}\}$, set of natural numbers $N = \{1, 2, 3, 4, \ldots\}$, set of integers $Z = \{\ldots, -3, -2, -1, 0, 1, 2, 3 \ldots\}$, set of rational numbers $Q = \left\{\frac{a}{b} : a \text{ and } b \text{ integer and } b \neq 0\right\}$, set of rational numbers R etc are infinite sets.

Example 4. Prove that the set of all natural numbers is an infinite set.

Solution: The set of natural numbers $N = \{1, 2, 3, 4, 5, 6, 7, 8, \ldots\}$

The set $A = \{1, 3, 5, 7, \ldots\}$ consisting of odd natural numbers from the set N

The set $B = \{2, 4, 6, 8, \ldots\}$ consisting of even natural numbers from the set N

The set of numbers $C = \{3, 6, 9, 12, \ldots\}$ multiple of 3 from N

Here, sets consisting of elements from N cannot be determined by counting. As a result A, B, C are infinite sets.

$\therefore$ N is an infinite set.

Work: Determine which ones are finite and which ones are infinite sets.

1) $\{3, 5, 7\}$
2) $\{1, 2, 2^2, \ldots 2^{10}\}$
3) $\{3, 3^2, 3^3, \ldots\}$
4) $\{x : x \text{ integer and } x < 4\}$
5) $\{\frac{p}{q} : p \text{ and } q \text{ are mutually coprimes and } q > 1\}$
6) $\{y : y \in N \text{ and } y^2 < 100 < y^3\}$

Empty Set

If a set does not have any element the set is said to be empty set, and is denoted by $\varnothing$. For example the set of three male students of Viqarunnissa school, $\{x \in N : 10 < x < 11\}$, $\{x \in N : x \text{ prime and } 23 < x < 29\}$ etc.

Venn-Diagram

John Venn (1834-1923) first expressed sets using pictures. Here every set is represented by a geometric figure like rectangles, cicrles and triangles in a plane. These pictures have been named after John Venn.

Subset

$A = \{a, b\}$ is a set. Sets $\{a, b\}$, $\{a\}$, $\{b\}$ can be formed from the elements of set A. Again, we can form a subset $\varnothing$ with no elements at all. These sets $\{a, b\}$, $\{a\}$, $\{b\}$, $\varnothing$ are subsets of set A. All the sets that can be formed from a set are called subsets of the original set. Subset is denoted by $\subseteq$. If B is a subset of A then we write $B \subseteq A$. B is a subset of A. Among the subsets above $\{a, b\}$ is equal to set A. Similarly, we can have a subset with no elements at all. $\therefore$ $\varnothing$ is a subset of any set.

Let $P = \{1, 2, 3\}$ and $Q = \{2, 3\}$, $R = \{1, 3\}$ then P, Q and R are subsets of P. That is, $P \subseteq P$, $Q \subseteq P$ and $R \subseteq P$.

Proper Subset

Of the subsets constructed from a set, the ones that have lesser number of elements than the original set is called proper set. For example, $A = \{3, 4, 5, 6\}$ and $B = \{3, 5\}$ are two sets. Here all elements of B are also elements of A, and number of elements in A is larger than that of B.

$\therefore$ B is a proper subset of A and is expressed as $B \subset A$.

In the examples of subsets Q and R are each proper subset of P. It may be mentioned here that $\varnothing$ is a proper subset of any set.

Example 5. Write down the subsets of $P = \{x, y, z\}$ and identify those that are proper subsets.

Solution: Given, $P = \{x, y, z\}$

Subsets of P are $\{x, y, z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x\}, \{y\}, \{z\}, \varnothing$.

Proper subsets of P are $\{x, y\}, \{x, z\}, \{y, z\}, \{x\}, \{y\}, \{z\}, \varnothing$

Observation: If number of elements of a set is n then it has 2^n subsets, and number of proper subsets is $2^n - 1$.

Equivalent Set

If two sets have the same elements then they are said to be equivalent sets. For example, $A = \{3, 5, 7\}$ and $B = \{5, 3, 3, 7\}$ are two equivalent sets, then they are denoted by $A = B$. Note that $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$

Again, if $A = \{3, 5, 7\}$, $B = \{5, 3, 3, 7\}$ and $C = \{7, 7, 3, 5, 5\}$ then sets A, B and C are equivalent sets. That is, $A = B = C$

Observation: If order of elements in the set are changed or certain element is repeated set is not changed.

Difference of Sets

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 5\}$. If we delete the elements of set B from that of A then we get $\{1, 2, 4\}$ and is denoted by $A \setminus B$ or $A - B$ and is read A delete B.

$\therefore A - B = \{1, 2, 3, 4, 5\} - \{3, 5\} = \{1, 2, 4\}$

Example 6. If $P = \{x : x$ is a factor of 12 $\}$ and $Q = \{x : x$ is a multiple of 3 and $x \leq 12\}$ then determine $P - Q$.

Solution: Given , $P = \{x : x \text{ is a factor of } 12\ \}$

Here, factors of 12 are $1, 2, 3, 4, 6, 12$

$\therefore P = \{1, 2, 3, 4, 6, 12\}$

Again, $Q = \{x : 3 \text{ is a factor of } x \text{ and } x \leq 12\}$

Here, numbers less than or equal to 12 that are divisible by 3 are $3, 6, 9, 12$

$\therefore Q = \{3, 6, 9, 12\}$

$\therefore P - Q = \{1, 2, 3, 4, 6, 12\} - \{3, 6, 9, 12\} = \{1, 2, 4\}$

The set is: $\{1, 2, 4\}$

Universal Set

All sets under discussion are subsets of a definite set. For example: the set $A = \{x, y\}$ is a subset of the set $B = \{x, y, z\}$ Here, B set is called as universal set with respect to set A.

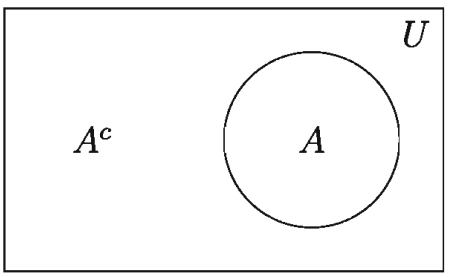
So if all sets under discussion are subsets of a given set then the later set is said to be **universal set** with respect to sets under discussion.

Usually universal set is denoted by U. However, other symbols can also be used to denote universal set.

For example, if set of all even natural numbers is $C = \{2, 4, 6, \ldots\}$ and set of all natural numbers is $N = \{1, 2, 3, 4, 5, 6, \ldots\}$ then N will be universal set with respect to the set C.

Complement of a Set

Let U be universal set and A be a subset of U. Then the set of elements outside of A is termed as **complementary set** of A. The complementary set of A is denoted by A^c or A'. Mathematically, $A^c = U \setminus A$.



Let, P and Q be two sets. Then the elements of P that are not elements of Q are said to form the complementary set of P with respect to Q, and is written $Q^c = P \setminus Q$

Example 7. If $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{2, 4, 6, 7\}$ and $B = \{1, 3, 5\}$ then determine A^c and B^c.

Solution: $A^c = U \setminus A = \{1,2,3,4,5,6,7\} \setminus \{2,4,6,7\} = \{1,3,5\}$

and $B^c = U \setminus B = \{1,2,3,4,5,6,7\} \setminus \{1,3,5\} = \{2,4,6,7\}$

The asked for sets are $A^c = \{1,3,5\}$ and $B^c = \{2,4,6,7\}$

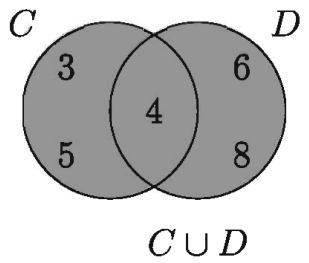
Union of Sets

The set consisting of elements of two or more sets is said to be **union of sets.** Let, A and B be two sets. The union of sets A and B is denoted by $A \cup B$ and read as A union B.

In set building method $A \cup B = \{x : x \in A \text{ or } x \in B\}$

Example 8. Let $C = \{3,4,5\}$ and $D = \{4,6,8\}$ determine $C \cup D$.

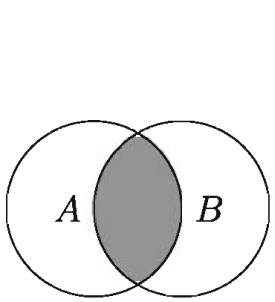
Solution: Given $C = \{3,4,5\}$
and $D = \{4,6,8\}$
$\therefore C \cup D = \{3,4,5\} \cup \{4,6,8\} = \{3,4,5,6,8\}$
The set to be determined is: $\{3,4,5,6,8\}$



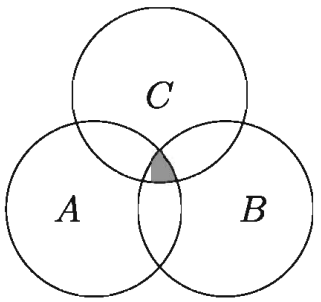
$C \cup D$

Intersection of Sets

The set consisting of common elements of two or more sets is known as **intersection set** Let A and B be two sets. The intersection of sets A and B is denoted by $A \cap B$ and read as A intersection B. In set building notation $A \cap B = \{x : x \in A \text{ and } x \in B\}$



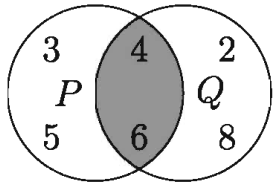
$A \cap B$



$A \cap B \cap C$

Example 9. If $P = \{x \in N : 2 < x \leq 6\}$ and $Q = \{x \in N : x \text{ is even and } x \leq 8\}$ determine $P \cap Q$.

Solution: Given
$P = \{x \in N : 2 < x \leq 6\} = \{3, 4, 5, 6\}$
$Q = \{x \in N : x \text{ is even and } x \leq 8\} = \{2, 4, 6, 8\}$
$\therefore P \cap Q = \{3, 4, 5, 6\} \cap \{2, 4, 6, 8\} = \{4, 6\}$
The set to be determined is $\{4, 6\}$



$P \cap Q$

Disjoint Set

If there is no element common to any two sets then these two sets are called **disjoint sets.**

Let A and B be two sets. If $A \cap B = \varnothing$ then A and B are said to be mutually disjoint sets.

Work: If $U = \{1, 3, 5, 9, 7, 11\}$, $E = \{1, 5, 9\}$ and $F = \{3, 7, 11\}$ then determine $E^c \cup F^c$ and $E^c \cap F^c$.

Power Sets

Let $A = \{m, n\}$ be a set. Subsets of A are $\{m, n\}, \{m\}, \{n\}, \varnothing$, and set of all sets $\{\{m, n\}, \{m\}, \{n\}, \varnothing\}$ of A is called A **power set** of A. Power set of A is denoted by $P(A)$. So the set formed by all subsets of a set is called power set of a given set.

Example 10. Find the number of elements of the power sets $A = \varnothing$, $B = \{a\}$, $C = \{a, b\}$.

Solution: Here $P(A) = \{\varnothing\}$

$\therefore$ number of elements of A is 0 and number of elements of its power set is $= 1 = 2^0$

Again $P(B) = \{\{a\}, \varnothing\}$

$\therefore$ number of elements of B is 1, and number of eleemnts of its power set is $= 2 = 2^1$.

and $P(C) = \{\{a\}, \{b\}, \{a, b\}, \varnothing\}$

$\therefore$ number of elements of C is 2 and number of elements of its power set is $= 4 = 2^2$

So if number of elements of a set is n, then number of elements of its power set is 2^n.

Work: If $G = \{1, 2, 3\}$ then determine $P(G)$. Show that number of elements of $P(G)$ is 2^3

Ordered Pair

Amena and Sumena of class VIII occupied 1st and 2nd position repectively in the annual examination. According to merit this can be expressed as (Amena, Sumena) pair. Similar definite pairs are called ordered pairs.

So for a pair of elements if the positions are given and expressed then the pair is called **Ordered Pair.**

If the first element of an ordered pair is x, and the second element is y then the ordered pair is denoted by (x, y). For the ordered pairs (x, y) and (a, b) $(x, y) = (a, b)$ if and only if $x = a$ and $y = b$.

Example 11. If $(2x + y, 3) = (6, x - y)$ then determine (x, y).

Solution: Given $(2x + y, 3) = (6, x - y)$

As per conditions of ordered pairs,

$2x + y = 6 \cdots\cdots (1)$

$x - y = 3 \cdots\cdots (2)$

By adding equations (1) and (2) we get $3x = 9$ or $x = 3$

Putting value of x in equation (1) we get, $6 + y = 6$ or $y = 0$

$\therefore (x, y) = (3, 0)$

(Cartesian Product)

Mr Karim decided to colour interior wall of his rooms using white or blue colour, and outside wall using red, yellow or green. The set of colours of interior wall is $A = \{$white, blue$\}$ and that of outside wall is $B = \{$red, yellow and green $\}$ Mr Karim can colour walls of his rooms by (white, red), (while, yellow), (white, green), (blue, red), (blue, yellow), (blue, green) ordered pairs. The above ordered pairs are writtena s below:

$A \times B = \{$(white, red), (while, yellow), (white, green), (blue, red), (blue, yellow), (blue, green)$\}$

The above set of ordered pairs is called **Cartesian Product set.**

In set building method $A \times B = \{(x, y) : x \in A \text{ and } y \in B\}$

$A \times B$ is read as A cross B.

Example 12. If $P = \{1, 2, 3\}$, $Q = \{3, 4\}$, $R = P \cap Q$ then determine $P \times R$ and $R \times Q$.

Solution: Given $P = \{1, 2, 3\}$, $Q = \{3, 4\}$

and $R = P \cap Q = \{1, 2, 3\} \cap \{3, 4\} = \{3\}$

$\therefore P \times R = \{1, 2, 3\} \times \{3\} = \{(1, 3), (2, 3), (3, 3)\}$

and $R \times Q = \{3\} \times \{3, 4\} = \{(3, 3), (3, 4)\}$

Work:

1) If $\left(\frac{x}{2} + \frac{y}{3}, 1\right) = \left(1, \frac{x}{3} + \frac{y}{2}\right)$ then determine (x, y).
2) If $P = \{1, 2, 3\}$, $Q = \{3, 4\}$ and $R = \{x, y\}$ then determine $(P \cup Q) \times R$ and $(P \cap Q) \times Q$.

Example 13. Determine the set of natural numbers that have the same residue 23 when they divide 311 and 419.

Solution: The natural numbers that leave 23 as residue when they divide 311 and 419 will be bigger than 23 and will be common divisor of both $311 - 23 = 288$ and $419 - 23 = 396$.

Let the set of natural numbers that are larger than 23 and that divide 288 be A

Here $288 = 1 \times 288 = 2 \times 144 = 3 \times 96 = 4 \times 72 = 6 \times 48 = 8 \times 36 = 9 \times 32 = 12 \times 24 = 16 \times 18$

$\therefore A = \{24, 32, 36, 48, 72, 96, 144, 288\}$

Let the set of natural numbers greater than 23 and that divided 396 is B

Here $396 = 1 \times 396 = 2 \times 198 = 3 \times 132 = 4 \times 99 = 6 \times 66 = 9 \times 44 = 11 \times 36 = 12 \times 33 = 18 \times 22$

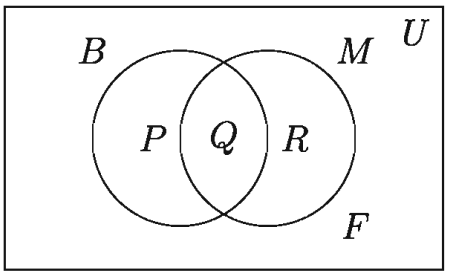
$\therefore B = \{33, 36, 44, 66, 99, 132, 198, 396\}$

$\therefore A \cap B = \{24, 32, 36, 48, 72, 96, 144, 288\} \cap \{33, 36, 44, 66, 99, 132, 198, 396\}$

$\therefore A \cap B = \{36\}$

The searched for set is $\{36\}$

Example 14. Of 100 students 88 passed in Bangla, 80 in Mathematics and 70 in both subjects. Express the information in Venn diagram, and find how many students failed in both subjects.



Solution: In the Venn diagram rectangular area represents U the set of 100 students. The set of students who passed in Bangla and Mathematics are denoted respectively by B and M. As a result the Venn diagram is divided into 4 disjoint sets denoted by P, Q, R and F.

Here, the set of students passed in both subjects is $Q = B \cap M$, cardinality of which is 70.

P is the set of students passed in Bangla only. Its cardinality is $|P| = 88 - 70 = 18$.

R is the set of students who passed only in Mathmatics. $\therefore$, $|R| = 80 - 70 = 10$

Students who passed in at least one subject is $P \cup Q \cup R = B \cup M$, and corresponding cardinality is $18 + 10 + 70 = 98$

F is the set of students who failed in both the subjects, and $|F| = 100 - 98 = 2$

$\therefore$ number of students failed in both the subjects is2.

Exercises 2.1

1. Express the following sets in set builder notation:
   1) $\{x \in N : x^2 > 9 \text{ and } x^3 < 130\}$
   2) $\{x \in Z : x^2 > 5 \text{ and } x^3 \leq 36\}$

3) $\{x \in N : x \text{ is factor of } 36 \text{ and } x \text{ is a multiple of } 6\}$

4) $\{x \in N : x^3 > 25 \text{ and } x^4 < 264\}$

2. Express the following sets in set builder notation:

1) $\{3, 5, 7, 9, 11\}$

2) $\{1, 2, 3, 4, 6, 9, 12, 18, 36\}$

3) $\{4, 8, 12, 16, 20, 24, 28, 32, 36, 40\}$

4) $\{\pm 4, \pm 5, \pm 6\}$

3. If $A = \{2, 3, 4\}$, $B = \{1, 2, a\}$ and $C = \{2, a, b\}$ then determine the following sets:

1) $B \setminus C$ 2) $A \cup B$ 3) $A \cap C$

4) $A \cup (B \cap C)$ 5) $A \cap (B \cup C)$

4. If $U = \{1, 2, 3, 4, 5, 6, 7\}$, $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$ and $C = \{3, 4, 5, 6, 7\}$ verify the following statements:

1) $(A \cup B)' = A' \cap B'$

2) $(B \cap C)' = B' \cup C'$

3) $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$

4) $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

5. If $Q = \{x, y\}$ and $R = \{m, n, l\}$ then determine $P(Q)$ and $P(R)$.

6. If $A = \{a, b\}$, $B = \{a, b, c\}$ and $C = A \cup B$ then show that number of elements of $P(C)$ is 2^n, where n is the number of elements of C.

7. 1) If $(x - 1, y + 2) = (y - 2, 2x + 1)$ then determine x and y.

2) If $(ax - cy, a^2 - c^2) = (0, ay - cx)$ then determine (x, y).

3) If $(6x - y, 13) = (1, 3x + 2y)$ then determine (x, y).

8. 1) If $P = \{a\}$, $Q = \{b, c\}$ then determine $P \times Q$ and $Q \times P$.

2) If $A = \{3, 4, 5\}$, $B = \{4, 5, 6\}$ and $C = \{x, y\}$ determine $(A \cap B) \times C$.

3) If $P = \{3, 5, 7\}$, $Q = \{5, 7\}$ and $R = P \setminus Q$ determine $(P \cup Q) \times R$.

9. If A and B are respectively sets of numbers that divide both 35 and 45, then determine $A \cup B$ and $A \cap B$.

10. Determine the natural numbers that divide both 346 and 556 to have 31 as residue.

11. Of the 30 students of a class 20 like football and 15 cricket. Number of students who like both the games is 10. Determine using Venn diagram the number of students who do not like either of the games.

12. Of 100 students 65 passed in Bangla, 48 in both Bangla and English and 15 failed in both the subjects.

    1) Express the above in Venn diagram.

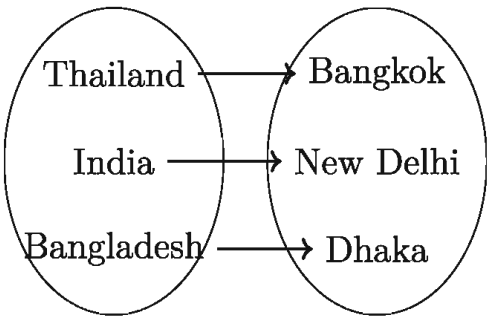
    2) Determine the number of students who passed only in Bangla or passed only in English.

    3) Determine union of the the sets of prime factors of number of students passed and failed in both the subjects.

Relations

We know Dhaka is the capital of Bangladesh, New Delhi is of India and Bangkok is of Thailand.

There is a relation between capitals and countries. This is country-capital relation. This relation can be expressed as a set in the following way:



That is country-capital relation = {(Bangladesh, Dhaka), (India, New Delhi), (Thailand, Bangkok)}.

If A and B are two sets then nonnull subset R of ordered pairs of their Cartesian product $A \times B$ is said to be a relation from the set A to the set B. Here set R is a subset of the set $A \times B$. That is, $R \subseteq A \times B$

Example 15. Let us assume that $A = \{3, 5\}$ and $B = \{2, 4\}$

$\therefore A \times B = \{3, 5\} \times \{2, 4\} = \{(3, 2), (3, 4), (5, 2), (5, 4)\}$

$\therefore R \subseteq \{(3, 2), (3, 4), (5, 2), (5, 4)\}$

when for an element x of set A and an element y of set B $(x, y) \in R$ then it is written $x\ R\ y$ and read that x is related to y, that is x is related to y using relation R.

If $x > y$ then $R = \{(3, 2), (5, 2), (5, 4)\}$

and if $x < y$ then $R = \{(3, 4)\}$

Again, if there is a relation from set A to set A, that is $R \subseteq A \times A$ then R is a relation of A.

If $x \in A$ $y \in B$ is related then the nonempty subset of ordered pairs (x, y) is called **relation.**

Example 16. If $P = \{2, 3, 4\}$, $Q = \{4, 6\}$ and for elements of P and Q a relation $y = 2x$ exists then find the relations.

Solution: Given $P = \{2, 3, 4\}$ and $Q = \{4, 6\}$

According to the question $R = \{(x, y) : x \in P, y \in Q \text{ and } y = 2x\}$

Here, $P \times Q = \{2, 3, 4\} \times \{4, 6\} = \{(2, 4), (2, 6), (3, 4), (3, 6), (4, 4), (4, 6)\}$

$\therefore R = \{(2, 4), (3, 6)\}$

The sought after relation is $\{(2, 4), (3, 6)\}$

Example 17. If $A = \{1, 2, 3\}$, $B = \{0, 2, 4\}$ and elements of A and B have the relation $x = y - 1$ find the relations.

Solution: Given $A = \{1, 2, 3\}$, $B = \{0, 2, 4\}$

According to the question relation $R = \{(x, y) : x \in A, y \in B \text{ and } x = y-1\}$

Here, $A \times B = \{1, 2, 3\} \times \{0, 2, 4\}$

$= \{(1, 0), (1, 2), (1, 4), (2, 0), (2, 2), (2, 4), (3, 0), (3, 2), (3, 4)\}$

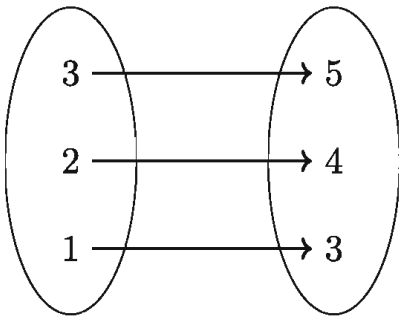
$\therefore R = \{(1, 2), (3, 4)\}$

Work: If $C = \{2, 5, 6\}$, $D = \{4, 5\}$ and the relation $x \leq y$ holds between elements of C and D find the relations.

Functions

Let us consider the relation between A and B:

For the relation $y = x + 2$
if $x = 1$ then $y = 3$
if $x = 2$ then $y = 4$
if $x = 3$ then $y = 5$



That is there is exactly one value for y for each value of x and x and y are related by $y = x + 2$. So two variables x and y are related in such a way that for each value of x there is exactly one value of y. Then y is said to be a **function** of x. Usually function of x is denoted by y, $f(x)$, $g(x)$, $F(x)$.

Let $y = x^2 - 2x + 3$ be a function. Here, for each value of x there is exactly one value of y. Here, both x and y are variables. Although both of them are variables value of y depends upon the value of x. That is why x is said to be **independent variable** and y **dependent variable**

Example 18. If $f(x) = x^2 - 4x + 3$ then determine $f(-1)$.

Solution: Given $f(x) = x^2 - 4x + 3$

$\therefore f(-1) = (-1)^2 - 4(-1) + 3 = 1 + 4 + 3 = 8$

Example 19. If $g(x) = x^3 + ax^2 - 3x - 6$ then for what value of a will $g(-2) = 0$?

Solution: Given $g(x) = x^3 + ax^2 - 3x - 6$

$\therefore g(-2) = (-2)^3 + a(-2)^2 - 3(-2) - 6$

$= -8 + 4a + 6 - 6 = 4a - 8$

As per question $g(-2) = 0$

$\therefore 4a - 8 = 0$ or $4a = 8$ or $a = 2$

$\therefore$ if $a = 2$ then $g(-2) = 0$.

Domain and Range

For ordered pairs of any relation the set of first elements is said to be **domain** and the set of second elements is said to be **range**.

Let R be a relation from set A to the set B, that is, $R \subseteq A \times B$. For the ordered pairs of R the set of first elements will be the domain of R and the set of second elements will be called range of R. The domain of R is denoted by *Dom* R and range by *Range* R.

Example 20. Find domain and range of the relation $S = \{(2,1),(2,2),(3,2),(4,5)\}$.

Solution: Given $S = \{(2,1),(2,2),(3,2),(4,5)\}$

The first elements of the ordered pairs of S are $2,2,3,4$ and the second elements are $1,2,2,5$

$\therefore$, *Dom* $S = \{2,3,4\}$ and *Range* $S = \{1,2,5\}$

Example 21. If $A = \{0,1,2,3\}$ and $R = \{(x,y) : x \in A, y \in A \text{ and } y = x+1\}$ then express R in tabular method and determine *Dom* R and *Range* R.

Solution: Given $R = \{(x,y) : x \in A, y \in A \quad y = x+1\}$

From the conditions of R we get $y = x+1$.

Now for each $x \in A$ determine the value of $y = x+1$.

x	0	1	2	3
y	1	2	3	4

Since $4 \notin A$, $(3,4) \notin R$. $\therefore R = \{(0,1),(1,2),(2,3)\}$

$\therefore$ Dom $R = \{0,1,2\}$ and Range $R = \{1,2,3\}$

Work:

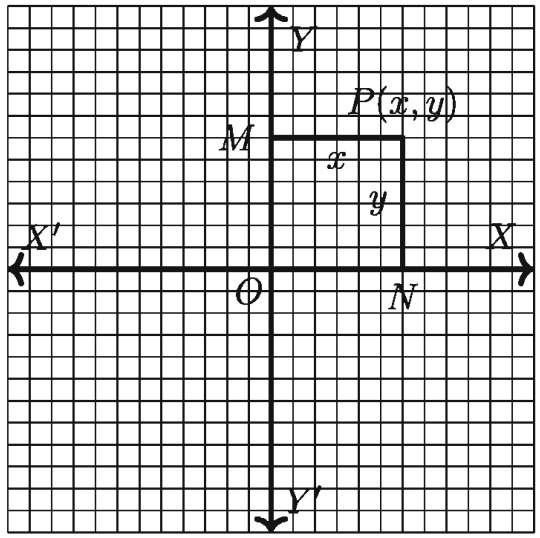
1) If $S = \{(-3,8),(-2,3),(-1,0),(0,-1),(1,0),(2,3)\}$ then determine domain and range of S.
2) Let $S = \{(x,y) : x,y \in A \text{ and } y - x = 1\}$, where $A = \{-3,-2,-1,0\}$. Determine Dom S and Range of S.

Graph of a Function

Visual display of a function is said to be **graph**. In order to clarify the concept of a function significance of graph is important. French philosopher and mathematician Rene Descartes (1596-1650) was the first to play pioneering role in introducing relations between algebra and geometry. He introduced a modern concept in geometry by defining the position of a point in two dimensional space by drawing two straight lines perpendicularly crossing to each other. He termed the perpendicular lines as axes, and their point of intersection as origin. Let two perpendicularly crossing straight lines XOX' and YOY' be drawn on a plane. Any point lying on this plane can be uniquely determined through these two

straight lines. Each of these perpendicular straight lines is called **axis**. The line XOX' parallel to horizon is called x-**axis**, and vertical line YOY' is called y-**axis** and point O of intersection of these lines is called **Origin**.

The signed perpendicular distance of aA point, located in the plane determined by two axes, from the axes are called **coordinates** of the point. Let P be any point on the plane determined by two axes. Let us draw perpendiculars PN and PM from P to XOX' and YOY'. Then $PM = ON$ which is perpendicular distance to YOY' from P and $PN = OM$ which is perpendicular distance from P to XOX'. If $PM = x$ and $PN = y$, then coordinates of point P are (x, y).



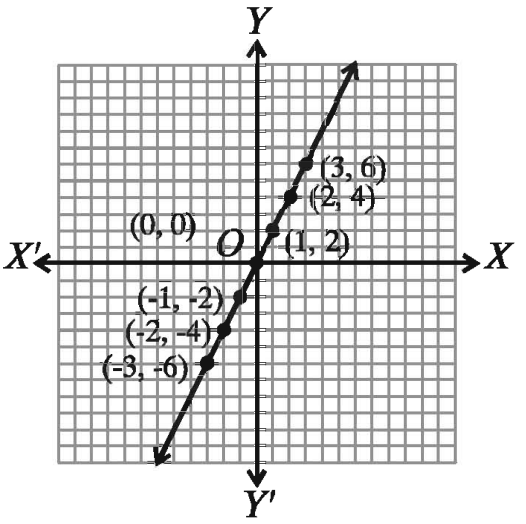
Here x is called **abscissa** and y is called **ordinate**. These coordinates are called **Cartesian coordinates.** It is very easy to depict a function using Cartesian coordinate system. Usually values of independent variable is set along x axis, and those of dependent variable along y axis.

In order draw a graph for the function $y = f(x)$ some values of independent variable are chosen from domain, and with corresponding values of dependent variables are formed ordered pairs. After that these ordered pairs are placed on the plane. The resultant points are then connected with line which is known as graph of the function $y = f(x)$.

Example 22. Draw the graph for the function $y = 2x$, where $-3 \leq x \leq 3$

Solution: Calculate values of y for some values of x from the domain $-3 \leq x \leq 3$, and draw a list.

x	-3	-2	-1	0	1	2	3
y	-6	-4	-2	0	2	4	6



In the graph paper each considering side of each smallest square as of length points of the list are marked nad added. This results in the graph of the function.

Example 23. If $f(y) = \dfrac{y^3 - 3y^2 + 1}{y(1-y)}$ then show that $f\left(\dfrac{1}{y}\right) = f(1-y)$

Solution: $f(y) = \dfrac{y^3 - 3y^2 + 1}{y(1-y)}$

$$\therefore f\left(\frac{1}{y}\right) = \frac{\left(\frac{1}{y}\right)^3 - 3\left(\frac{1}{y}\right)^2 + 1}{\frac{1}{y}\left(1 - \frac{1}{y}\right)} = \frac{\frac{1 - 3y + y^3}{y^3}}{\frac{y-1}{y^2}}$$

$$= \frac{1 - 3y + y^3}{y^3} \times \frac{y^2}{y-1} = \frac{1 - 3y + y^3}{y(y-1)}$$

Again, $f(1-y) = \dfrac{(1-y)^3 - 3(1-y)^2 + 1}{(1-y)(1-(1-y))}$

$$= \frac{1 - 3y + 3y^2 - y^3 - 3(1 - 2y + y^2) + 1}{(1-y)(1-1+y)}$$

$$= \frac{1 - 3y + 3y^2 - y^3 - 3 + 6y - 3y^2 + 1}{y(1-y)}$$

$$= \frac{-1 + 3y - y^3}{y(1-y)} = \frac{-(1 - 3y + y^3)}{-y(y-1)}$$

$$= \frac{1 - 3y + y^3}{y(y-1)}$$

$\therefore f\left(\dfrac{1}{y}\right) = f(1-y).$

Example 24. Let Universal set be $U = \{x : x \in N \text{ and } x \leq 6\}$, $A = \{x : x$ prime and $x \leq 5\}$, $B = \{x : x$ even number and $x \leq 6\}$ and $C = A \setminus B$

1) Determine A^c.
2) Prove that, $A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$
3) Prove that, $(A \cap C) \times B = (A \times B) \cap (C \times B)$

Solution:

1) Given $U = \{x : x \in N \text{ and } x \leq 6\} = \{1, 2, 3, 4, 5, 6\}$

   $A = \{x : x \text{ prime and } x \leq 5\} = \{2, 3, 5\}$

   $\therefore\ A^c = U \setminus A = \{1, 2, 3, 4, 5, 6\} - \{2, 3, 5\} = \{1, 4, 6\}$

2) Given

   $B = \{x : x \text{ prime and } x \leq 6\} = \{2, 4, 6\}$

   $\therefore\ A \cup B = \{2, 3, 5\} \cup \{2, 4, 6\} = \{2, 3, 4, 5, 6\} \cdots\cdots (1)$

   $A \setminus B = \{2, 3, 5\} - \{2, 4, 6\} = \{3, 5\}$

   $B \setminus A = \{2, 4, 6\} - \{2, 3, 5\} = \{4, 6\}$

   $A \cap B = \{2, 3, 5\} \cap \{2, 4, 6\} = \{2\}$

   $\therefore\ (A \setminus B) \cup (B \setminus A) \cup (A \cap B) = \{3, 5\} \cup \{4, 6\} \cup \{2\} = \{2, 3, 4, 5, 6\} \cdots (2)$

   Therefore comparing (1) and (2) we get,

   $A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$

3) From (2) we get,

   $C = A \setminus B = \{3, 5\}$

   $A \cap C = \{2, 3, 5\} \cap \{3, 5\} = \{3, 5\}$

   $\therefore (A \cap C) \times B = \{3, 5\} \times \{2, 4, 6\}$

   $= \{(3, 2), (3, 4), (3, 6), (5, 2), (5, 4), (5, 6)\} \cdots\cdots (3)$

   $A \times B = \{2, 3, 5\} \times \{2, 4, 6\}$

   $= \{(2, 2), (2, 4), (2, 6), (3, 2), (3, 4), (3, 6), (5, 2), (5, 4), (5, 6)\}$

   $C \times B = \{3, 5\} \times \{2, 4, 6\}$

   $= \{(3, 2), (3, 4), (3, 6), (5, 2), (5, 4), (5, 6)\}$

   $\therefore\ (A \times B) \cap (C \times B)$

$$= \{(2,2),(2,4),(2,6),(3,2),(3,4),(3,6),(5,2),(5,4),(5,6)\}$$
$$\cap \{(3,2),(3,4),(3,6),(5,2),(5,4),(5,6)\}$$
$$= \{(3,2),(3,4),(3,6),(5,2),(5,4),(5,6)\} \cdots\cdots (4)$$

Therefore, comparing (3) and (4),

$(A \cap C) \times B = (A \times B) \cap (C \times B)$

Example 25. Let $A = \{4,5,6,7\}$, $B = \{0,1,2,3\}$ and $R = \{(x,y) : x \in A, y \in A \text{ and } y = x+1\}$

1) Prove that, sets A and B are disjoint.
2) Determine $P(B)$ to show that number of elements of $P(B)$ is 2^n, where n is the number of elements of B.
3) Express the relation R in tabular method and determine domain.

Solution:

1) Given, $A = \{4,5,6,7\}$ and $B = \{0,1,2,3\}$

   $\therefore\ A \cap B = \{4,5,6,7\} \cap \{0,1,2,3\} = \varnothing$

   Since $A \cap B = \varnothing$

   A and B are disjoint.

2) Given,

   $B = \{0,1,2,3\}$

   $\therefore\ P(B) = \{\{0\},\{1\},\{2\},\{3\},\{0,1\},\{0,2\},\{0.3\},\{1,2\},\{1,3\},\{2,3\},$
   $\{0,1,2\},\{0,1,3\},\{0,2,3\},\{1,2,3\},\{0,1,2,3\},\varnothing\}$

   Here number of elements of B is 4 and number of elements of the power set is $2^4 = 16$

   $\therefore$ if n is the number of elements of B then number of elements of power set is 2^n.

   $\therefore$ number of elements of $P(B)$ satisfies 2^n formula.

3) Given, $R = \{(x,y) : x \in A, y \in A \text{ and } y = x+1\}$ and $A = \{4,5,6,7\}$

   From the conditions of R we get $y = x+1$

   Now, for each $x \in A$ draw a list by finding values of $y = x+1$.

x	4	5	6	7
y	5	6	7	8

Since $8 \notin A$, therefore $(7,8) \notin R$

$\therefore\ R = \{(4,5),(5,6),(6,7)\}$

$Dom\ R = \{4,5,6\}$

Exercises 2.2

1. Which one is the set of factors of 8?
   1) $\{8,16,24,\cdots\}$ 2) $\{1,2,4,8\}$
   3) $\{2,4,8\}$ 4) $\{1,2\}$
2. If R is a relation from the set C to the set B, then which one of the following is true ?
   1) $R \subset C$ 2) $R \subset B$ 3) $R \subseteq C \times B$ 4) $C \times B \subseteq R$
3. If $A = \{1,2\}, B = \{2,5\}$ which one of the following is the number of elements of $P(A \cap B)$?
   1) 1 2) 2 3) 3 4) 8
4. Which of the following expresses the set $\{x \in N : 13 < x < 17$ and x is prime $\}$ in tabular method?
   1) $\varnothing$ 2) $\{0\}$ 3) $\{\varnothing\}$ 4) $\{13,17)\}$
5. If $A \cup B = \{a,b,c\}$ then
   (i) $A = \{a,b\},\ B = \{a,b,c\}$
   (ii) $A = \{a,b,c\},\ B = \{b,c\}$
   (iii) $A = \{a,b\},\ B = \{c\}$
   Based upon the above facts which one of the following holds?
   1) i 2) ii 3) i and ii 4) i, ii and iii
6. If for two finite sets A and B
   (i) $A \times B = \{(x,y) : x \in A$ and $y \in B\}$
   (ii) If $n(A) = a, n(B) = b$ then $n(A \times B) = ab$
   (iii) Each member of $A \times B$ is an ordered pair.
   Based upon the statements above which one of the following is true?

1) *i* and *ii* 2) *i* and *iii* 3) *ii* and *iii* 4) *i*, *ii* and *iii*

If $A = \{6, 7, 8, 9, 10, 11, 12, 13\}$ then answer the questions 7–9 below:

7. Which one is the correct expression for the set A?
   1) $\{x \in N : 6 < x < 13\}$ 2) $\{x \in N : 6 \leq x < 13\}$
   3) $\{x \in N : 6 \leq x \leq 13\}$ 4) $\{x \in N : 6 < x \leq 13\}$
8. Which one is the set of primes in A?
   1) $\{6, 8, 10, 12\}$ 2) $\{7, 9, 11, 13\}$ 3) $\{7, 11, 13\}$ 4) $\{9, 12\}$
9. Which one of the following sets is a multiple of 3 in set A ?
   1) $\{6, 9\}$ 2) $\{6, 11\}$ 3) $\{9, 12\}$ 4) $\{6, 9, 12\}$
10. If $A = \{3, 4\}, B = \{2, 4\}$, $x \in A$ and $y \in B$ then determine the relation $x > y$ in A and B.
11. If $C = \{2, 5\}, D = \{4, 6, 7\}$, $x \in C$ and $y \in D$ then determine the relation $x + 1 < y$ in C and D.
12. If $f(x) = x^4 + 5x - 3$ then determine $f(-1), f(2)$ and $f\left(\frac{1}{2}\right)$.
13. If $f(y) = y^3 + ky^2 - 4y - 8$, then for what value of k will $f(-2) = 0$?
14. If $f(x) = x^3 - 6x^2 + 11x - 6$ then for what value of x is $f(x) = 0$?
15. If $f(x) = \dfrac{2x+1}{2x-1}$ then determine the value of $\dfrac{f\left(\frac{1}{x^2}\right) + 1}{f\left(\frac{1}{x^2}\right) - 1}$.
16. If $g(x) = \dfrac{1 + x^2 + x^4}{x^2}$ prove that $g\left(\frac{1}{x^2}\right) = g(x^2)$
17. Determine domain and range from the following relations.
    1) $R = \{(2, 1), (2, 2), (2, 3)\}$
    2) $S = \{(-2, -4), (-1, 1), (0, 0), (1, 1), (2, 4)\}$
    3) $F = \{(\frac{1}{2}, 0), (1, 1), (1, -1), (\frac{5}{2}, 2), (\frac{5}{2}, -2)\}$
18. Express the following relations in tabular method and determine domain and range for each.
    1) $R = \{(x, y) : x \in A, y \in A \text{ and } x + y = 1\}$ where $A = \{-2, -1, 0, 1, 2\}$
    2) $F = \{(x, y) : x \in C, y \in C \text{ and } y = 2x\}$ where $C = \{-1, 0, 1, 2, 3\}$

19. Draw the points $(-3, 2), (0, -5), \left(\frac{1}{2}, -\frac{5}{6}\right)$ in graph paper.

20. Draw the points $(1, 2), (-1, 1), (11, 7)$ on the graph paper and show that all the three points are on the same straight line.

21. Universal set $U = \{x : x \in N \text{ and } x \text{ odd number }\}$

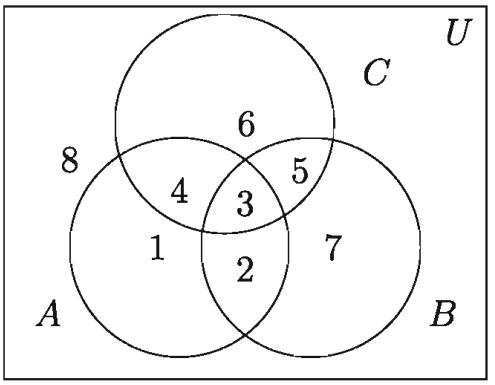
    $A = \{x : x \in N \text{ and } 2 \leq x \leq 7\}$

    $B = \{x : x \in N \text{ and } 3 < x < 6\}$

    $C = \{x : x \in N \text{ and } x^2 > 5 \text{ and } x^3 < 130\}$

    1) Express set A in tabular method.
    2) Determine A' and $C \setminus B$.
    3) Determine $B \times C$ and $P(A \cap C)$.

22. Look at the Venn diagram.



    1) Express the set B using set building method.
    2) Using the facts mentioned above verify the relation $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
    3) If $S = (B \cup C)^c \times A$ then determine dom S.

23. $y = f(x) = \frac{4x - 7}{2x - 4}$ is a function.

    1) Determine the value of $f\left(-\frac{1}{2}\right)$.
    2) Determine the value of $\frac{f(x) + 2}{f(x) - 1}$.
    3) Prove that $f(y) = x$